

SYLLABUS :-

AR 23203: BUILDING CONSTRUCTION SESSIONAL I Credits: 2 L-T-P: 0-0-3

Prerequisite: None Course Overview: The course intends to familiarize architecture students with basic constructional practices prevalent in conventional buildings. Based on the lectures delivered and reference drawings provided, the students shall be required to draw construction drawings and details. These lectures will be supplemented by relevant visual aids and site visits. The students will be encouraged to draw constructional details popular in current practices while being acquainted with traditional construction practices as well. The understanding of this course will act as a prerequisite for the next semester course Building Construction Sessional - II and Working Drawing I and II in the subsequent semesters. Learning Objectives: 1.

Understand and illustrate sectional details of a conventional brick construction building at foundation, plinth, sill, lintel, terrace and parapet levels. 2. Recognize the various classifications of doors and windows based on mode of operation, material used and its composite parts. 3. Get acquainted with the fixing details of external façade cladding using different materials like stone, brick, composite panels etc. 4. Understand the constructional details of various types of staircases and design a staircase for a given hall. 5. Examine the different types and parts of pitched roofs while enumerating and illustrating various types of timber and steel roof trusses and roof coverings.

Course Curriculum: Module 1 : Skin section of conventional brick buildings Through foundation, plinth, window or door sill, lintel with chajja, roof slab, cornice and parapet details etc. Module 2 : Doors and windows Different types of doors and windows made of various materials like Timber, MS, Aluminium etc. Module 3 : Cladding Different types of cladding: stone, brick, glazed tiles, earthen tiles, plastic tiles, composite panels etc. Module 4 : Staircase Different type of staircases, sectional details along with railing details, both guard rail and staircase railing with fixing details. Module 5 : Roof Trusses Different types of pitched roofs and nomenclature of its members and parts. Timber and steel trusses, types; fixing details showing purlin, rafter, tie, strut, cleat etc. Different types of roof coverings: tiles, asbestos and metal sheets etc. with fixing details Reading List: 1. Arora, S.P. Bindra, S.P. (2010). A Textbook of Building Construction, New Delhi: Dhanpat Rai Publications. 2. Bureau of Indian Standard (BIS), (2016). National Building Code of India 2016 (Volume-1 2). 3. Ching, F. D. K. (1991). Building Construction Illustrated. New York, NY: Van Nostrand Reinhold. 4. Chudley, R. Greeno, R. (2008). Building Construction Handbook. Oxford: Elsevier Ytytytyt 5. Emmitt, S. Gorse, C. (2010). Barry's Introduction to Construction of Buildings. New Jersey: Wiley Blackwell. 6. Frederick, S.M. Jonathan, T. R. (2000). Building Design and Construction Handbook, McGraw Hill 7. Gatz, K. (2000). Modern Architectural Detailing. New York: Van Nostrand Reinhold. 8. Mckay, W.B. (2004). Building Construction, (Volume 1 to 5), Routledge 9.

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